

Transparency between Research Methods and Results

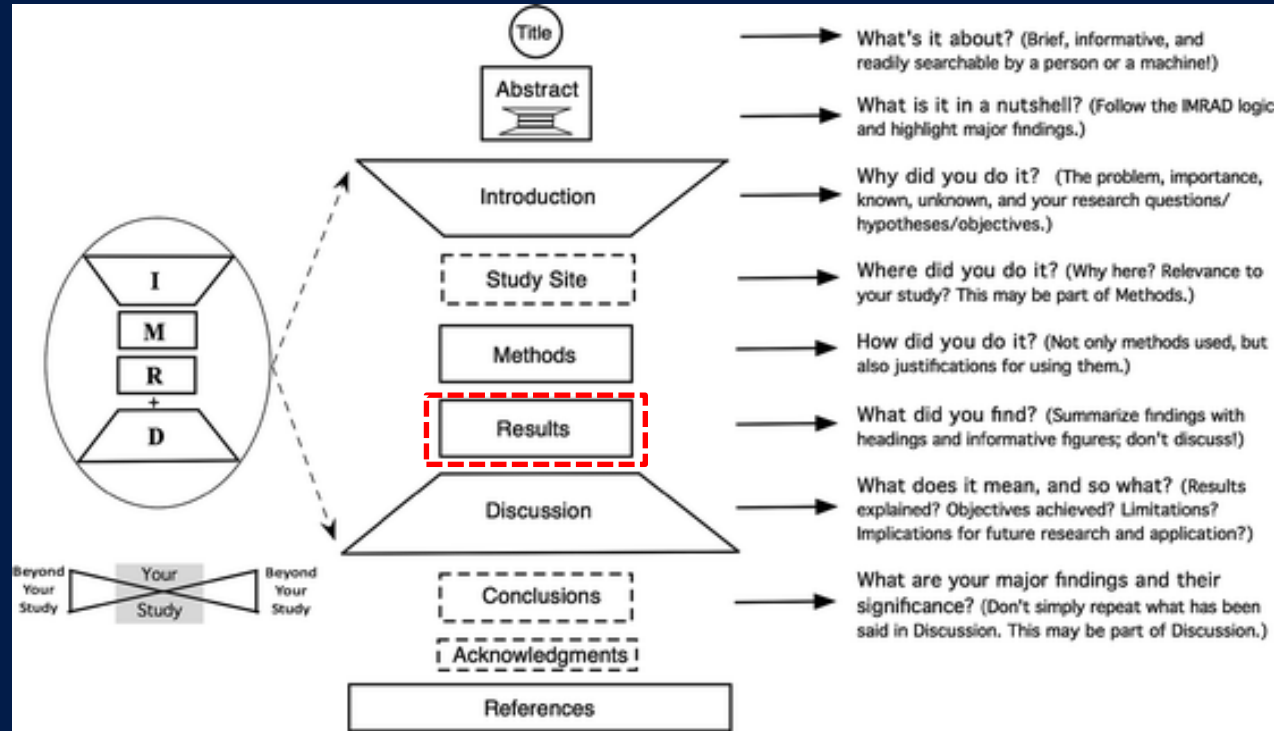
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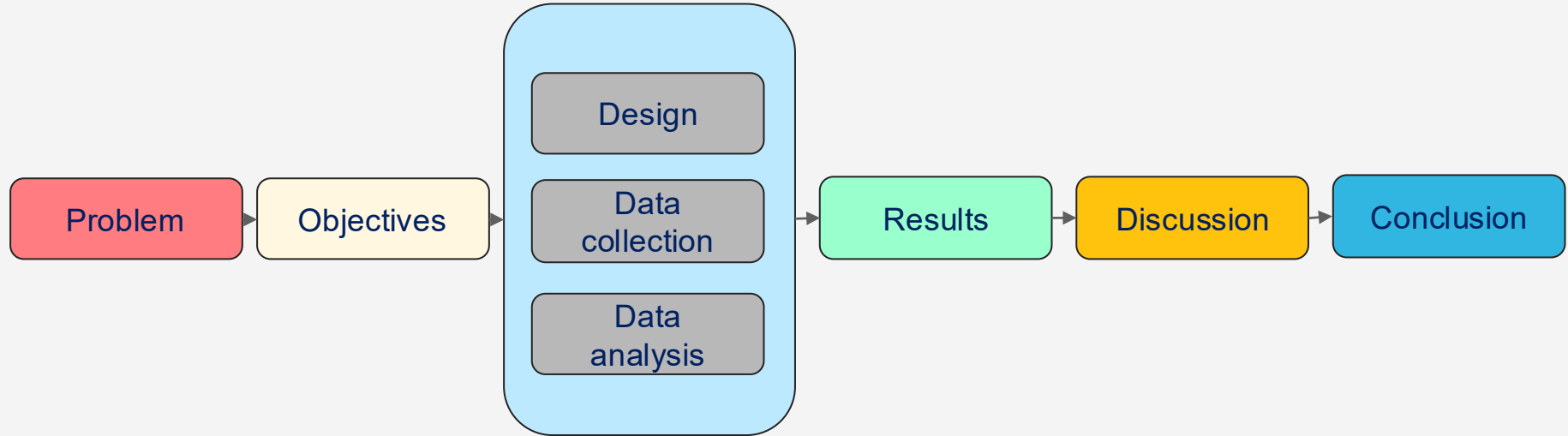
Outline

- Defining transparency in research
- Linking methods to results
- Reporting quantitative and qualitative results transparently

Recap on IMRAD method



Research Workflow



Transparency

methodological transparency refers to “the degree of detail and disclosure about the specific steps, decisions, and judgment calls made during a scientific study”

Transparency

**Entails clear reporting of design,
sampling, analysis, uncertainty,
limitation and ethics**

Benefits of research transparency

- Credibility
- Reproducibility
- Scientific Integrity
- Trustworthiness

Objectives → Methods → Results

Objective	Data Collection	Analysis	Results
To determine flood exposure	Household survey	Descriptive statistics	Exposure percentages
To understand community perceptions	Interviews	Thematic analysis	Themes
To predict flood risk	GIS & Remote Sensing	Random Forest	Flood risk maps

Discuss more examples from participants

**No results should be
reported without
corresponding method**

Reporting results from quantitative methods

Present findings in the same order as described in Methods.

Descriptive Statistics

- Means
- Standard deviations
- Frequencies
- Percentages

Inferential Statistics

Include

- Test statistic
- Degrees of freedom
- p-values
- Confidence intervals
- Effect size

Tables, Figures and Maps

Good visualizations should include:

- Clear titles
- Units
- Legends
- Sample size
- Source
- Numbering

Using Maps to report results

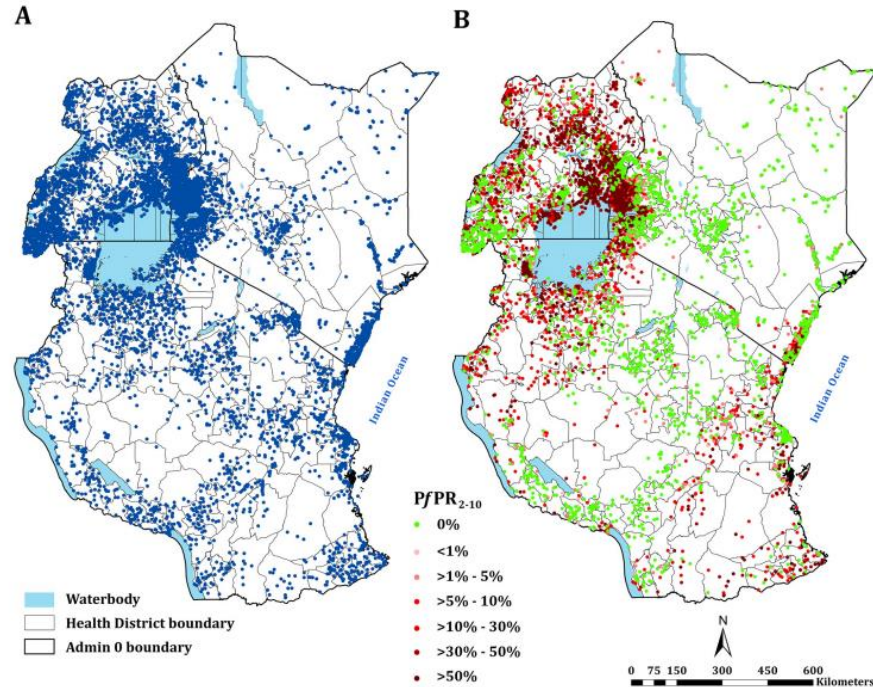


Fig 1. Assembled parasite rate surveys. (A) Distribution of all assembled survey data ($n = 18940$) between 2010-2020; (B) the distribution of age-corrected and microscopy-standard parasite prevalence ($PfPR_{2-10}$) estimates among samples ≥ 10 individuals with the highest values on top when multiple surveys conducted at the same location. Base shapefiles used in all figures downloaded from: Kenya- <https://data.humdata.org/dataset/ken-administrative-boundaries>; Uganda- <https://data.humdata.org/dataset/uganda-administrative-boundaries-admin-1-admin-3> and Tanzania- <https://data.humdata.org/dataset/tanzania-administrative-boundaries-level-1-to-3-regions-districts-and-wards-with-2012-population> <https://gadm.org/>.

<https://doi.org/10.1371/journal.pgph.0000014.g001>

Using Charts to report results

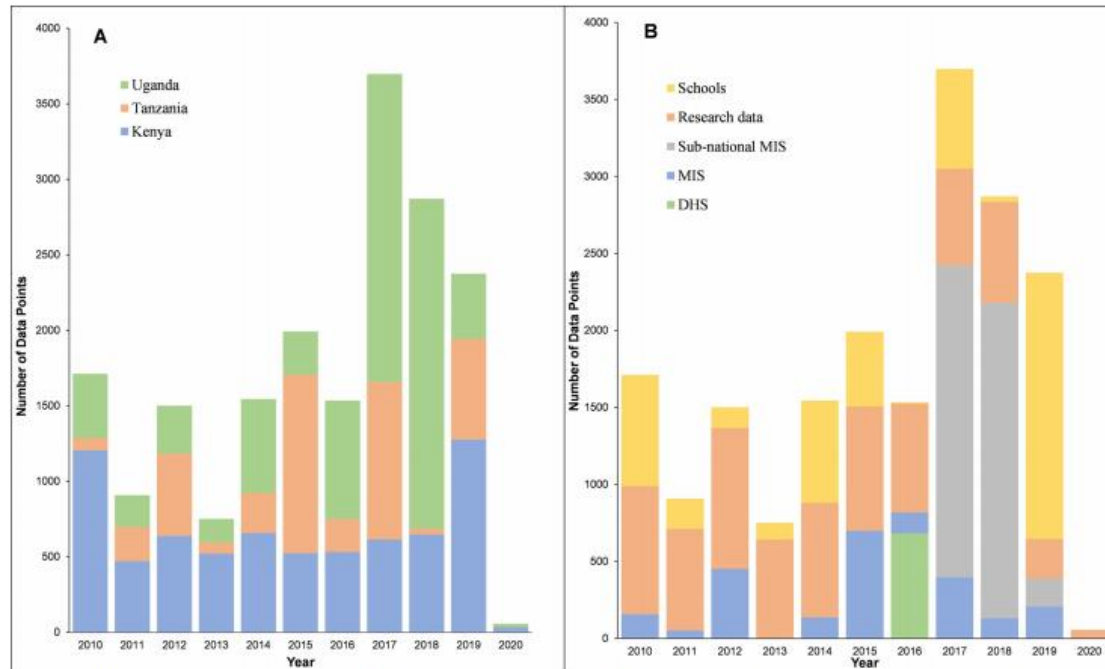


Fig 2. Assembled surveys by year and data source. (A) Temporal distribution of surveys 2010–2020 by country (B) Temporal distribution of surveys 2010–2020 according to the data source

Reporting results from qualitative methods

Present findings by theme

Theme 1: Limited Access to Climate Information

Quotation

"Flood warnings usually arrive after the flooding has already started."

Interpretation

Delayed communication increases household vulnerability.

Reporting results from mixed methods

Integrate quantitative and qualitative findings

Quantitative	Qualitative	Interpretation
72% flooded	Residents cited blocked drainage	Infrastructure contributes to flood risk
63% lacked insurance	Participants reported poverty	Financial vulnerability limits resilience

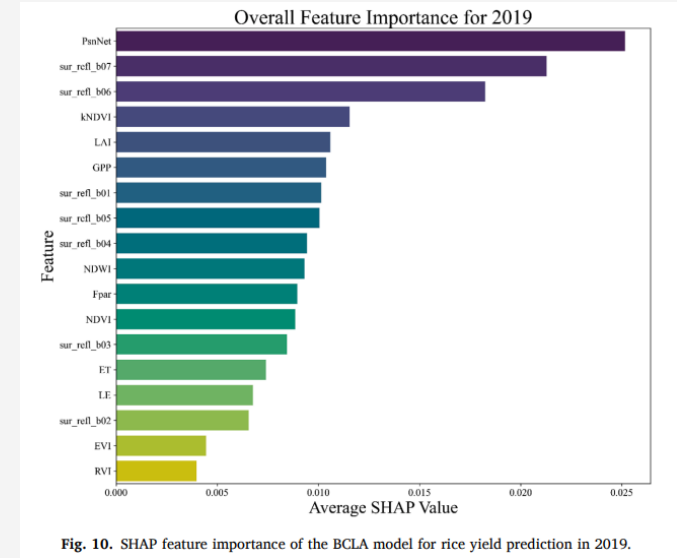
Reporting results from AI and GIS

For GIS studies report

- Spatial resolution
- Coordinate system
- Classification algorithm
- Accuracy assessment
- Validation

For AI studies report

- Model used
- Hyperparameters
- Feature selection
- Accuracy
- Precision
- Recall
- F1-score
- Explainability



Mistakes

- Reporting results not described in Methods
- Introducing new variables
- Mixing Results with Discussion
- Too many quotations without analysis
- Poor figure captions
- Lack of uncertainty reporting

Best Practices

- Every objective should have a corresponding method and results
- Results should be presented in the same order of the methods
- Use tables and figures effectively
- Report uncertainty and limitations of the study

Questions and discussions